serviceOrchestrationPlantManagement

Service Description

Service ID: *"serviceOrchestrationPlantManagement"*

Abstract
This document describes the `serviceOrchestrationPlantManagement` service, version 1.0.0, produced by the DynamicServiceOrchestration core system of Arrowhead 5.2.2.

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1 Overview

This document describes the `serviceOrchestrationPlantManagement` service, which provides management of the active engineered plant snapshot used by DynamicServiceOrchestration. It is produced by the DynamicServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

Local 2026-10-23 extension replacing the PlantDescriptionEngine dependency on SimpleStore.

1.2 How This Service Is Meant to Be Used

PDE queries and atomically replaces the complete list of consumer/service scopes and ranked provider instance IDs. Lower nonnegative priority wins among otherwise eligible providers. LocalCloudManager may query, but only PDE or Sysop may replace, in addition to the common management policy.

1.3 Important Delimitations and Dependencies

HTTP only. Empty provider lists intentionally deny unrelated alternatives, including inter-cloud fallback. An empty rules list clears the active plant. DSO still enforces authorization, blacklist, locks, interface and QoS constraints. Existing consumer caches may delay visible changes. Apply database_patch_pde_dso.sql before starting an upgraded DSO.

2 Service Interface

This section describes the interfaces to the `serviceOrchestrationPlantManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.



Figure 1: SysML-style block of the `serviceOrchestrationPlantManagement` service and its operations.

The following interface operations are available.

2.1 operation query () : OrchestrationPlantRules

Returns the current active-plant snapshot; allowed for authenticated PDE, LocalCloudManager or Sysop, also subject to the common management policy.

2.2 operation replace (OrchestrationPlantRules) : OrchestrationPlantRules

Atomically validates and replaces the complete snapshot. Only authenticated PDE or Sysop may replace; invalid input does not modify existing state. Empty rules clears the active plant; empty providers retains an unavailable managed scope.

3 Information Model

Here, all data objects that can be part of the `serviceOrchestrationPlantManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.4, which are used to represent things like hashes and identifiers.

3.1 struct OrchestrationPlantProvider

Allowed service instance and its engineered priority.

Field	Type	Description
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
priority	Number	Nonnegative rank; zero is highest and lower values win among otherwise eligible providers.

3.2 struct OrchestrationPlantRule

Managed consumer/service scope, retained even without an available provider.

Field	Type	Description
consumer	Name	Name of the consumer system.
serviceDefinition	Name	Service definition name.
providers	List<OrchestrationPlantProvider>	Ranked allowed service instances; empty means managed but unavailable, with no unrelated or inter-cloud fallback.

3.3 struct OrchestrationPlantRules

Atomic snapshot owned by PlantDescriptionEngine at DSO.

Field	Type	Description
rules	List<OrchestrationPlantRule>	Complete active-plant snapshot; an empty array clears all managed scopes.

3.4 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.

4 References

[1] J. Delsing, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-SysD.pdf.

[2] ——, “serviceOrchestrationPlantManagement HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, dynamicServiceOrchestration-v5.2.2-serviceOrchestrationPlantManagement-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	